Smart OPD Queue and Appointment Management System

Software Requirements Specification (SRS)

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1. Introduction

1.1 Purpose of this Document

This document defines the product requirements for the Smart OPD Queue and Appointment Management System. It describes the system's intended behaviour, users, interfaces, performance targets, constraints, quality attributes, and verification approach. It is intended to guide product design, implementation, testing, project planning, and stakeholder review.

The document is written as a buildable product specification. Requirements use "shall" for mandatory behaviour. Numeric targets are initial engineering targets and should be confirmed before production deployment.

1.2 Scope of this Document

The system manages outpatient appointments and the live queue around them. Rather than treating an appointment token as a guaranteed fixed consultation time, it maintains a current queue state and estimates when each patient is likely to be seen. The estimate is refreshed when relevant events occur, such as a consultation finishing, a doctor running late, a patient failing to arrive, a break being recorded, or an authorised staff member adding an emergency case.

The first release includes patient registration/sign-in, department and doctor selection, appointment booking, token issuance, dynamic estimated time of arrival/consultation (ETA), queue tracking, arrival alerts, check-in, doctor queue controls, configurable no-show handling, staff-controlled emergency prioritisation, workload visibility, and optional suggestions for less-loaded clinically appropriate doctors or slots.

The project cannot rely on an on-ground hospital trial for its initial evaluation. Therefore, the SRS includes a randomized, discrete-event simulation environment that generates synthetic OPD days and evaluates queue/ETA behaviour reproducibly. Simulation is a pre-deployment engineering check; it is not evidence that the product has already improved outcomes in a real hospital.

Out of scope for the first release: diagnosis or clinical decision-making; automated determination of medical urgency; electronic medical records or clinical notes; billing/payments; replacing a hospital information system; and automatically moving a patient to another doctor without authorised confirmation.

1.3 Overview

The product has three operational parts: (1) patient, doctor, and administrator interfaces; (2) a queue-management service that records authoritative appointment and queue states; and (3) an ETA service that recalculates expected consultation times from the current queue and observed operational events. A separate simulation harness generates synthetic events and feeds them through the same queue/ETA logic used by the product, without connecting to real patients or altering live queues.

The intended value is practical: patients receive more useful arrival guidance, front-desk staff have a consistent way to handle no-shows and exceptions, doctors can operate the queue, and administrators can see workload and congestion. An ETA is always an estimate, not a promise.

1.4 Intended Audience

Product owner, hospital/OPD stakeholders, developers, UI/UX designers, testers, project managers, and future maintenance/support staff.

1.5 Definitions and Requirement Language

OPD: Outpatient Department. ETA: estimated time at which a patient is expected to be attended; it may change. Token: unique queue reference. Check-in: confirmation that the patient has arrived and is ready to be called. No-show: a patient who has not checked in by the configured cutoff. Grace period: additional time allowed before the system marks a patient as eligible to be skipped. Queue event: a status change that can affect queue order or ETA. Discrete-event simulation: a model that advances through events such as arrivals, consultation starts/ends, breaks, and emergencies rather than simulating every second.

FR = functional requirement; PR = performance requirement; NFR = non-functional requirement; SIM = simulation requirement; AC = acceptance criterion.

2. General Description

2.1 Product Perspective

The system is a standalone OPD queue and appointment platform for its first release. It may later integrate with hospital information systems or messaging providers, but such integrations are not assumed to exist in this version. The queue state stored by the application is the authoritative operational record for the features in scope.

2.2 Product Functions

Patients can register/sign in, select a department or doctor, book an available appointment, receive a token and initial ETA, view queue progress, receive arrival/change alerts, and check in.

Doctors or authorised OPD staff can view the relevant queue, call the next eligible patient, mark a consultation as started/completed, record delays or breaks, and skip a patient under policy.

Administrators can manage departments, doctors, OPD hours, queue rules, access roles, emergency actions, and workload views.

The ETA service estimates consultation start times and refreshes them when material queue events occur. The system records meaningful actions so a queue change can be explained and audited.

The simulation environment creates synthetic patients and operating conditions, executes many repeatable OPD-day scenarios, compares the product's queue/ETA logic with a baseline, and produces metric summaries.

2.3 User Classes and Characteristics

Patient: Needs clear booking details, token, current ETA, queue status, and understandable alerts. The patient must not be able to change queue order or see another patient's private information.

Doctor: Needs a low-friction view of the current queue and controls for calling, starting, completing, and skipping consultations.

Front-desk / OPD staff: Needs check-in and patient-status controls, no-show handling, and the ability to communicate or escalate exceptions within assigned permissions.

Administrator: Needs to configure departments, doctor schedules, OPD timings, queue policies, user roles, and workload dashboards.

Developer / Tester: Needs a deterministic simulation interface, scenario configuration, repeatable random seeds, structured logs, and exportable test results.

2.4 Operating Environment

The initial product is assumed to be a responsive web application accessible from common desktop and mobile browsers. A server-side application and persistent database are required. A notification provider and hosting environment must be selected before deployment. No specialised hardware is required.

The simulation must run in a development/test environment and must not send real patient notifications or write to production queue data.

2.5 Assumptions and Dependencies

The hospital or test operator maintains correct doctor availability, OPD hours, queue policy, and patient status.

System timestamps are sufficiently consistent to order queue events. A configured notification provider is available for real alert delivery; simulation can use a mock provider.

Hospital stakeholders approve who may mark emergencies, skip or reinsert patients, and change queue order.

Because no on-ground test is available during initial development, synthetic scenarios and documented assumptions will be used. Any later real-world deployment still requires operational, privacy, and safety review.

2.6 Product Decisions and Default Rules

Arrival alert: by default, notify a checked-in/expected patient when the current ETA is approximately 30 minutes away. This threshold must be configurable by OPD.

No-show grace period: default to 10 minutes after the applicable call/arrival cutoff, configurable by OPD. The system must not silently delete a missing patient.

Late arrival/reinsertion: default to the end of the eligible queue after check-in, unless an authorised staff member applies a permitted hospital policy exception. Reinsertion must be logged.

Emergency priority: emergency insertion is a staff-controlled action. The software does not assess clinical urgency. The authorised action triggers ETA recalculation for affected queue entries.

ETA refresh: recalculate after a material event, including consultation completion, start, delay, break, skip, reinsertion, emergency insertion, doctor availability change, or a relevant no-show state transition.

ETA messaging: show the time as an estimate and explain that it may change. Do not imply a guaranteed consultation time.

Load balancing: suggestions are advisory only and must respect department/doctor suitability rules. A suggestion does not change an appointment unless the patient or authorised staff confirms it.

3. Functional Requirements

3.1 Accounts, Roles, and Patient Details

- **FR-001:** The system shall allow a patient to create an account or sign in before using account-based booking features.
- **FR-002:** The system shall authenticate users and apply role-based permissions for patient, doctor, staff, and administrator functions.
- **FR-003:** The system shall collect only the patient details needed for booking, contact, identification, and the chosen notification channel.
- **FR-004:** A patient shall be able to view only their own appointment and queue details. Staff shall see patient information only as needed for their role.

3.2 Appointment Booking and Token

- **FR-005:** The patient shall be able to select a department and, where applicable, a doctor from currently available options.
- **FR-006:** The system shall prevent booking against unavailable doctors, closed OPD hours, or invalid appointment options.
- **FR-007:** On successful booking, the system shall create a unique appointment reference and queue token, record the booking time, and show the initial ETA.
- **FR-008:** The system shall show the appointment's doctor/department, date, token, current status, and ETA together.
- **FR-009:** The system shall handle duplicate or conflicting booking submissions without creating duplicate active appointments.

3.3 Queue State and Dynamic ETA

- **FR-010:** The system shall maintain a single authoritative queue state for each active doctor/OPD queue.
- **FR-011:** The ETA service shall estimate the expected consultation start time using available queue position, observed consultation durations, doctor availability, recorded breaks/delays, and applicable queue rules.
- **FR-012:** The system shall recalculate ETAs after each material queue event. The list of material events shall include consultation start/completion, delay, break, skip, reinsertion, emergency insertion, no-show transition, and doctor availability change.
- **FR-013:** The patient, doctor, and administrator views shall reflect the same latest queue state after an update has been committed.

- **FR-014:** The interface shall identify the ETA as an estimate and shall show when it was last updated.
- **FR-015:** If an ETA cannot be calculated reliably, the system shall display an appropriate unavailable/temporarily delayed status rather than a fabricated precise time.
- **FR-016:** The system shall retain enough queue-event history to reproduce the order of events used to calculate an ETA and investigate discrepancies.

3.4 Arrival Alerts and Notifications

- **FR-017:** The system shall issue an arrival alert when the configured arrival threshold is reached and a valid contact/channel is available.
- **FR-018:** The system shall notify affected patients of material ETA changes when notification rules determine that the change warrants an alert.
- **FR-019:** Notification attempts shall record channel, purpose, timestamp, and delivery outcome where the provider exposes that outcome.
- **FR-020:** Notification failure shall not block booking, queue progression, or consultation controls. The interface shall show the alert as failed, pending, or unconfirmed where status is known.
- **FR-021:** The system shall prevent repeated duplicate alerts for the same event unless a retry policy explicitly allows them.

3.5 Check-in, Consultation, and No-Show Handling

- **FR-022:** An authorised user or supported patient workflow shall be able to record check-in with a timestamp.
- **FR-023:** Doctor or authorised staff shall be able to call the next eligible patient, start consultation, complete consultation, and skip a patient.
- **FR-024:** The system shall apply the configured no-show grace period and shall make a patient eligible to be skipped only according to the configured rule.
- **FR-025:** A skipped patient shall remain recorded with a skipped status and reason where supplied; the system shall not silently delete the appointment.
- **FR-026:** The system shall support reinsertion of a skipped patient under the configured policy, defaulting to the end of the eligible queue after check-in.
- **FR-027:** Each skip, reinsertion, and manual override shall record the actor, timestamp, action, and reason where required by policy.
- **FR-028:** Queue and ETA updates shall be triggered when these actions change the active queue.

3.6 Emergency-Aware Queue

- **FR-029:** An authorised staff member shall be able to mark an appointment/queue entry for emergency priority under the hospital's configured policy.
- **FR-030:** The system shall not infer, diagnose, or independently determine emergency priority.
- **FR-031:** After an emergency insertion, the system shall update queue order and recalculate affected ETAs according to the configured policy.
- **FR-032:** The system shall log the staff member who applied emergency priority and shall notify affected patients when notification rules and contact availability permit.

3.7 Workload Intelligence and Load Balancing

- **FR-033:** The administrator view shall show queue length, current consultation/patient state, average observed consultation duration, estimated queue-clearing time, and congestion indicators for relevant doctors.
- **FR-034:** The system shall identify less-loaded doctor/slot options only when they satisfy configured department and suitability constraints.

- **FR-035:** Load-balancing recommendations shall be advisory. The system shall not autonomously transfer a patient or override clinical judgement.
- **FR-036:** Administrators shall be able to configure doctor availability, department associations, OPD hours, arrival-alert threshold, no-show grace period, and permitted queue policies.

3.8 Simulation and Evaluation

- **SIM-001:** The system shall provide a separate simulation harness for generating synthetic OPD operating days without using identifiable real patient data.
- **SIM-002:** The simulator shall generate configurable synthetic patients, appointment arrivals, check-in behaviour, consultation durations, no-shows, doctor breaks, delays, emergency insertions, and notification failures.
- **SIM-003:** Each run shall accept and record a random seed so the same configuration and seed can reproduce the same synthetic event stream.
- **SIM-004:** The simulator shall feed generated events through the same queue-state and ETA logic used by the application, through a test adapter that cannot modify production data or send real notifications.
- **SIM-005:** The test harness shall support a baseline policy, such as a fixed-token/basic FIFO queue with no event-driven ETA refresh, and compare it with the Smart OPD policy under the same generated scenario.
- **SIM-006:** The harness shall support normal-operation, disruption, and stress scenarios, including combinations of no-shows, longer-than-expected consultations, doctor breaks, emergencies, and notification failures.
- **SIM-007:** Each experiment shall run multiple seeds per scenario. The initial target is at least 100 seeds per scenario, where compute resources permit, and the report shall state the actual run count.
- **SIM-008:** The harness shall calculate ETA mean absolute error (MAE), median absolute error, percentage of ETAs within ±15 minutes of actual simulated consultation start, physical waiting time after check-in, ETA revision frequency/magnitude, queue throughput, no-show/reinsertion outcomes, and notification success/failure counts.
- **SIM-009:** The report shall include scenario configuration, seed/run count, baseline and Smart OPD results, metric definitions, summary statistics, and confidence intervals or other uncertainty summaries where appropriate.
- **SIM-010:** Simulation results shall be labelled simulated. Reports shall not claim that they prove real-world effectiveness, clinical safety, or actual reductions in hospital waiting time.
- **SIM-011:** Simulation code and the queue/ETA engine shall be separable so that the simulator can be replaced or extended without rewriting core queue logic.

3.9 Administration and Audit

- **FR-037:** Administrators shall be able to create/update departments and doctor profiles and manage OPD schedules.
- **FR-038:** The system shall provide an operational overview of active queues, doctor workload, delays, and exceptions.
- **FR-039:** Material administrative actions and queue overrides shall be recorded in an audit log with actor, action, timestamp, affected object, and outcome.
- **FR-040:** The system shall provide a way to export simulation results and operational test summaries for review.

4. Interface Requirements

4.1 User Interfaces

Patient interface: registration/sign-in, department/doctor selection, booking, token, ETA, queue status, check-in, notification state, and clear messages when an ETA changes.

Doctor interface: current patient, queue list, call-next, start, complete, skip, and controls for recording a delay or break where authorised.

Staff interface: patient lookup/check-in, no-show handling, reinsertion, and permitted queue exception actions.

Administrator interface: department/doctor management, OPD timing and policy settings, queue monitoring, workload indicators, audit history, and simulation report access.

Simulation/test interface: scenario selection, configuration of arrival rates and event distributions, random seed, baseline selection, run controls, and metric/report output. This interface shall be restricted to test/development roles.

4.2 Software Interfaces

The application shall communicate with its persistent database through a defined data-access layer. The notification service shall be accessed through a provider adapter so the provider can be changed without rewriting queue logic.

No hospital information system, electronic health record, identity provider, or third-party messaging integration is assumed in the first release. Any such integration requires a separate interface specification and stakeholder approval.

The simulation shall use an event adapter and mock notification provider. It shall not point at production databases, production credentials, or real patient contact lists.

4.3 Communications

Client/server traffic shall use encrypted transport in deployed environments. Queue updates should be pushed to active interfaces where practical; polling may be used as a fallback.

Notification channels may include SMS, email, push notification, or a messaging provider selected for deployment. The chosen provider, costs, delivery guarantees, and consent requirements must be confirmed before release.

4.4 Hardware Interfaces

No specialised hardware is required. Supported phones, tablets, laptops, and desktop computers with a modern browser are assumed. Printers or physical token displays are optional future integrations and are not required for the first release.

5. Performance Requirements

5.1 Dynamic Performance Targets

- **PR-001:** Under the initial test load of 500 concurrent active sessions, 95% of ordinary page/API requests should complete within 2 seconds, excluding external notification-provider latency.
- **PR-002:** After a committed material queue event, 95% of affected ETA and queue views should reflect the recalculated state within 5 seconds under the initial test load.
- **PR-003:** The queue-state service shall prevent two conflicting queue actions from silently overwriting one another. Conflicting actions shall be rejected or resolved by a defined consistency rule.
- **PR-004:** The simulator shall support at least 100 synthetic OPD-day runs per scenario in batch mode, subject to the declared hardware/runtime budget; if not achieved, the report shall record the limitation.
- **PR-005:** The simulation report shall compute ETA metrics against simulated actual consultation start times, not against the ETA output itself.

5.2 Static Capacity Requirements

The application shall store unique identifiers for patients, appointments, queue entries, queue events, and simulation runs. The initial design shall not hard-code a single hospital, department, or doctor limit; expected production capacity shall be confirmed before deployment.

The system shall retain the timestamps and event records needed to calculate queue state, reconstruct a simulation run, and investigate operational discrepancies, subject to the retention policy approved by the deploying organisation.

5.3 ETA Quality and Evaluation

Initial simulation acceptance targets: (a) report MAE and median absolute error for every scenario; (b) report the percentage of ETAs within ±15 minutes; (c) report check-in-to-consultation waiting time and ETA instability; and (d) compare all key metrics with the baseline under identical generated scenarios.

The initial engineering target is for at least 80% of simulated ETAs to fall within ±15 minutes of the simulated consultation start in normal-operation scenarios. This is a target to evaluate, not a claim of achieved accuracy. Disruption and stress scenarios must be reported separately rather than hidden in an aggregate score.

The team shall investigate major regressions, unstable ETA oscillations, incorrect queue order, or worse results than baseline. Thresholds for disruption/stress scenarios shall be documented after initial simulator calibration.

6. Design Constraints

6.1 Clinical and Operational Constraints

The system shall not make diagnoses, decide clinical urgency, or replace staff judgement. Emergency priority, queue overrides, and load-balancing actions shall remain subject to authorised human control and hospital policy.

Queue policy must be configurable because OPD workflows vary. The default 30-minute arrival alert and 10-minute no-show grace period are starting values, not universal clinical rules.

The ETA service must communicate uncertainty honestly. It must not present an exact time as a guarantee when operational data is incomplete or stale.

6.2 Technical and Data Constraints

The queue-state and ETA logic shall be testable independently from the user interface.

The simulation shall be isolated from live data and external notification delivery. Synthetic records shall be clearly marked and must not be confused with real patients.

Technology stack, cloud vendor, database engine, notification provider, and deployment model are implementation decisions unless separately mandated by the project.

Patient data collection shall be minimised. Data retention, deletion, backup, access, and applicable legal/privacy requirements must be finalised before real deployment.

6.3 Verification Constraint: No On-Ground Test

The initial validation plan shall not depend on access to a hospital floor, real patients, or live clinical operations. The team shall use randomized synthetic data and discrete-event simulation to exercise the system under repeatable conditions.

Simulation inputs shall be configurable rather than presented as measured hospital facts. Distributions for consultation duration, arrival patterns, no-show rates, delay frequency, and emergency frequency shall be documented as assumptions until reliable operational data becomes available.

Passing simulation tests demonstrates that the implementation behaves as expected under the modelled conditions only. It does not establish that model assumptions match a real OPD. Before production use, the deploying organisation should review the workflow and model assumptions with qualified OPD stakeholders and perform an appropriately governed operational validation when feasible.

7. Non-Functional Attributes

7.1 Security and Privacy

- **NFR-001:** Users shall be authenticated before accessing protected features.
- **NFR-002:** Role-based access control shall restrict actions and patient data according to the user’s assigned role.

- **NFR-003:** Sensitive data shall be protected in transit and at rest using controls appropriate to the chosen deployment.
- **NFR-004:** The system shall not expose one patient's private appointment/contact details to another patient.
- **NFR-005:** The simulation shall use synthetic data by default and shall not require production patient identifiers or contact details.

7.2 Reliability and Recovery

- **NFR-006:** Appointment and queue state shall be persisted before the system reports a state-changing action as successful.
- **NFR-007:** A notification-provider outage shall not stop core queue progression. Failed delivery shall be recorded where possible and shall not be treated as proof that a patient received an alert.
- **NFR-008:** If ETA computation fails, the system shall preserve the last known queue state and clearly indicate that the ETA may be unavailable or stale.
- **NFR-009:** Queue state shall remain internally consistent after concurrent actions, retries, or recoverable application errors.

7.3 Usability and Accessibility

- **NFR-010:** Token, queue status, and ETA shall be easy to locate and use plain language.
- **NFR-011:** Material ETA changes shall be communicated clearly without implying certainty.
- **NFR-012:** Patient-facing pages shall be responsive on common mobile screen sizes and support keyboard navigation, visible focus, and accessible labels for interactive controls.
- **NFR-013:** Errors shall explain the action that failed and, where possible, the next step.

7.4 Maintainability, Testability, and Reproducibility

- **NFR-014:** Queue policy values such as alert threshold and no-show grace period shall be configurable rather than hard-coded.
- **NFR-015:** Queue management and ETA calculation shall have automated unit and integration tests.
- **NFR-016:** Simulation runs shall record software version, configuration, random seed, scenario identifier, run timestamp, and metrics.
- **NFR-017:** The same software version, scenario configuration, and random seed shall reproduce the same event sequence, within documented deterministic limits.
- **NFR-018:** Simulation reports shall include metric definitions and disclose omitted factors, assumptions, and limitations.

7.5 Scalability and Portability

- **NFR-019:** The application shall separate interface, business logic, persistence, notification adapters, and simulation tooling so components can be changed independently.
- **NFR-020:** The initial system shall run in a documented development environment and support repeatable setup and test execution. Production scaling targets shall be finalised after expected hospital volume is known.

8. Preliminary Schedule

8.1 Preliminary Delivery Schedule

Phase 1 — Requirements and architecture: 1 week: finalise workflow, role permissions, data model, event definitions, and interface wireframes.

Phase 2 — Core queue and booking: 2 weeks: accounts/roles, department and doctor setup, booking, token issuance, check-in, and queue state.

Phase 3 — ETA and queue operations: 2 weeks: ETA engine, event-driven recalculation, doctor controls, no-show policy, breaks/delays, and emergency insertion.

Phase 4 — Patient communication and dashboards: 1–2 weeks: arrival/change alerts, notification status, workload dashboard, and audit events.

Phase 5 — Simulation harness: 1-2 weeks: synthetic event generator, scenario configuration, random seeds, baseline comparison, metrics, and report export.

Phase 6 — Verification and hardening: 1 week: automated tests, performance checks, security review, accessibility checks, reproducibility checks, and documentation.

8.2 Schedule Estimate

Estimated duration: approximately 7–9 weeks for a small team working in parallel, assuming the project scope and team availability remain stable. This is a planning estimate, not a committed delivery date. Integration work, stakeholder review, and deployment approvals may extend the schedule.

8.3 Milestones

Milestone 1: approved requirements and architecture. Milestone 2: booking and queue workflow demonstrated. Milestone 3: ETA updates and exception handling demonstrated. Milestone 4: simulator produces reproducible baseline comparisons. Milestone 5: acceptance test report and release candidate completed.

9. Appendices

Appendix A — Simulation Plan

A1. Synthetic inputs: Generate synthetic patients, appointment schedules, arrival/check-in times, consultation durations, doctor schedules, breaks, late starts, no-shows, emergency insertions, and notification outcomes. Inputs should be configurable and distributions documented.

A2. Scenario families: Normal day: expected arrival and consultation patterns. Disruption day: longer consultations, late doctor starts, breaks, and clustered no-shows. Emergency day: one or more staff-entered emergency insertions. Notification-degradation day: delayed or failed alerts. Stress day: higher booking volume, concurrent state changes, and multiple disruptions.

A3. Comparison method: Run the baseline and Smart OPD policy against the same synthetic scenario and seed. Avoid comparing different random event streams, since that could make the comparison unfair. Report results per scenario as well as overall.

A4. Metrics: ETA MAE and median absolute error; share of ETAs within ±15 minutes; check-in-to-consultation waiting time; number and size of ETA revisions; throughput; no-show and reinsertion outcomes; notification attempts and failures; queue-order violations; and runtime.

A5. Repeatability and reporting: Record seed, configuration, scenario, software version, number of runs, summary statistics, and uncertainty intervals where appropriate. Save machine-readable raw results and a human-readable report.

A6. Limitations: Synthetic data is only as realistic as the assumptions used. Simulation cannot fully capture human behaviour, clinical judgement, hospital-specific exceptions, connectivity conditions, or organisational constraints. Results must be described as simulated outcomes.

Appendix B — Initial Acceptance Criteria

- **AC-001:** A valid booking produces one unique appointment reference, token, and initial ETA.
- **AC-002:** ETA is recalculated after consultation completion, delay/break, skip/reinsertion, doctor availability change, or authorised emergency insertion.
- **AC-003:** Patient, doctor, and administrator views converge on the same committed queue state.
- **AC-004:** Arrival alerts follow the configured threshold; notification failures do not block queue progression and are recorded where status is available.
- **AC-005:** The configured no-show grace period and reinsertion rules are applied and recorded.
- **AC-006:** Users cannot perform actions outside their assigned permissions.

- **AC-007:** Workload view includes queue length, current patient/consultation, average consultation duration, estimated clearing time, and congestion indicator.
- **AC-008:** Load-balancing suggestions do not automatically transfer patients or override clinical suitability.
- **AC-009:** The simulator runs a fixed scenario with a recorded seed and reproduces the event sequence under the same version/configuration.
- **AC-010:** Simulation output compares the baseline and Smart OPD policy on the required metrics and labels the results as simulated.
- **AC-011:** At least 100 seeds per scenario are attempted for the initial evaluation where resources permit; actual counts and any shortfall are disclosed.
- **AC-012:** Reports do not claim real-world effectiveness based solely on simulation.

Appendix C — Core Data Entities

Patient: identifier and minimum account/contact fields. Doctor: identifier, department, availability, and schedule. Department: identifier and name. Appointment: patient, doctor/department, booking time, token, status, and timestamps. Queue entry: appointment reference, position/priority, status, check-in time, and skip/reinsertion details. Consultation: start/end timestamps and status. ETA snapshot: estimate, calculation time, and queue-state version. Notification: purpose, channel, timestamp, and delivery status. Queue policy: OPD hours, grace period, alert threshold, and permitted queue rules. Audit event: actor, action, affected object, timestamp, and outcome. Simulation run: scenario, seed, software version, configuration, metrics, and result location.

Appendix D — Acronyms

ETA — Estimated Time of Arrival / estimated consultation time. OPD — Outpatient Department. SRS — Software Requirements Specification. MAE — Mean Absolute Error. FIFO — First In, First Out. UI — User Interface. API — Application Programming Interface.

Appendix E — Decisions Required Before Production

Confirm supported deployment platform and hosting; notification channel and associated cost; identity and authentication approach; hospital-specific emergency and no-show policy; data retention/deletion and applicable privacy requirements; expected patient and concurrent-user volume; operational support and availability targets; and whether future hospital-system integrations are required.

10. Conclusion

10.1 Summary

The Smart OPD Queue and Appointment Management System is intended to make OPD visits more predictable by maintaining a live queue, recalculating ETAs when operational events occur, supporting consistent no-show and emergency workflows, and giving staff visibility into workload. It supports staff decisions rather than replacing clinical judgement.

Because on-ground testing is unavailable during initial development, the product includes a formal simulation-based verification approach. Repeatable synthetic scenarios and baseline comparisons can expose implementation defects and evaluate behaviour under stated assumptions. They cannot prove real-world impact; model assumptions and acceptance results must be reported transparently, and later deployment should include appropriate operational review.